Taibah University
College of Computer Science and Engineering
Department of Cybersecurity

CYB 270 – Cryptography

Project

"Text File Encryption and Decryption Tool"

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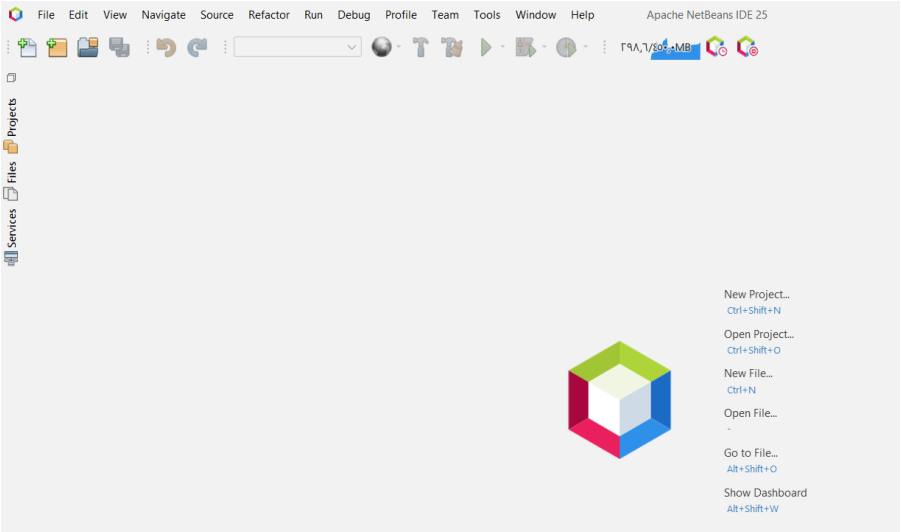
1. Introduction

In the digital world, cryptography is essential for protecting data. The main objective of this project is to provide a clear and practical introduction to the fundamental concepts of encryption and decryption. We developed a simple tool that lets users experiment with text encryption and decryption using the DES, Caesar Cipher, and Morse Code algorithms. By choosing an algorithm, entering plaintext, and then seeing the ciphertext that is produced (and vice versa), users can directly learn about both substitution-based techniques and more intricate block ciphers. This tool teaches how encryption works in practice and demonstrates how it can be both simple and complex. [1] [2]

2. Getting Started

2.1 System Requirements

1. Operating system such as Windows.
2. An appropriate JDK and JavaFX SDK (if not within JDK) for the operating system.
3. IDE such as netbeans to run the tool.



2.2 Installation Steps

1. make sure to download and install JDK and JavaFX SDK on your device.
2. set up the selected IDE and run the tool program.
3. A GUI will appear to help users easily encrypt or decrypt messages.

3. How to Use the Tool

3.1 Interface Description

The application's GUI is developed using **JavaFX** and written in NetBeans IDE. Upon launching the tool, users are presented with a basic and easy-to-use interface with two primary modes:

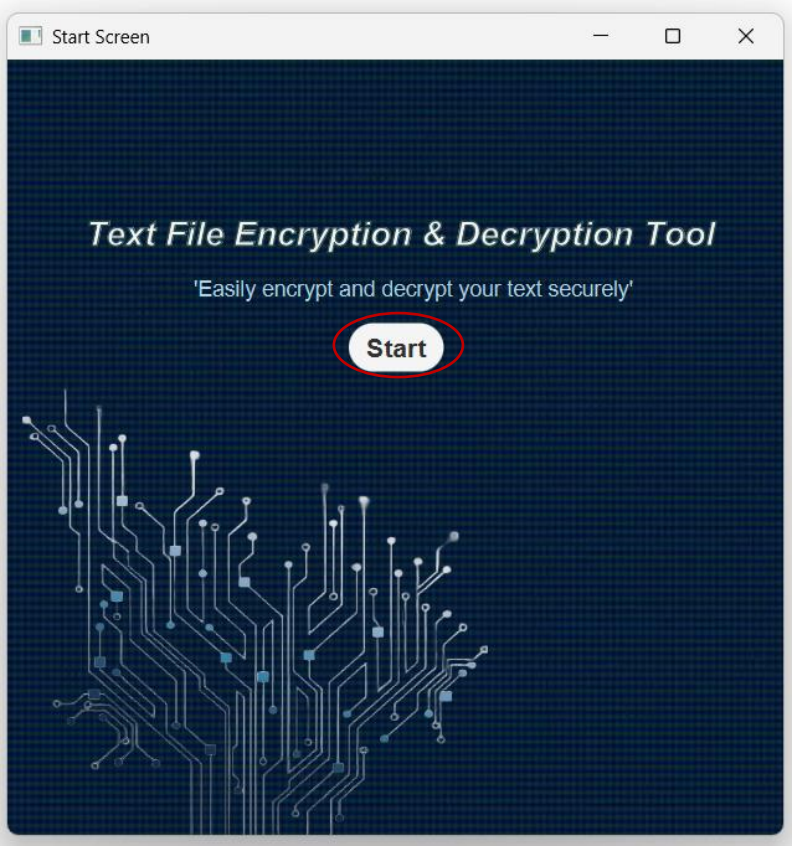
Encryption Mode: Users can enter a text file, choose an encryption algorithm, enter an encryption key (if required), and generate encrypted output.

Decryption Mode: A user uploads an encrypted file or write cipher text, selects the appropriate algorithm, enters the proper decryption key, and receives the plain text.

Step-by-Step Guide to Using the Tool

Step 1: Start the Tool

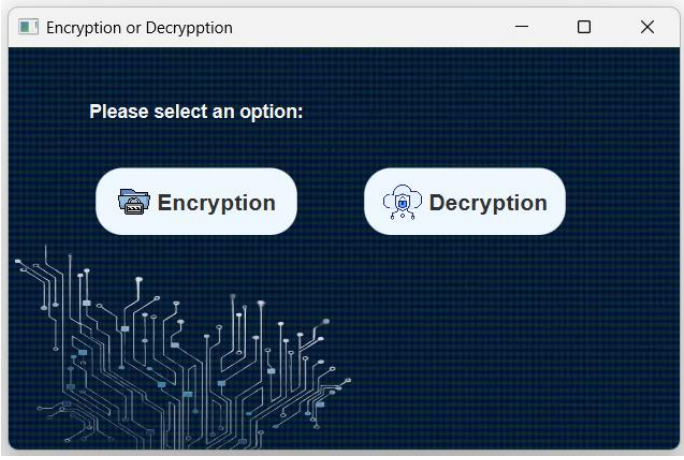
- ❖ Launch the application.
- ❖ "On the start screen, click the **'Start'** button located at the center of the window.



Step 2: Choose a Mode

❖ After clicking the **"Start"** button, a new screen will appear showing two options:

- **Encryption**
- **Decryption**



❖ Select one of the modes based on your task:
- If you want to **encrypt** a file, proceed to **Step 3A**.

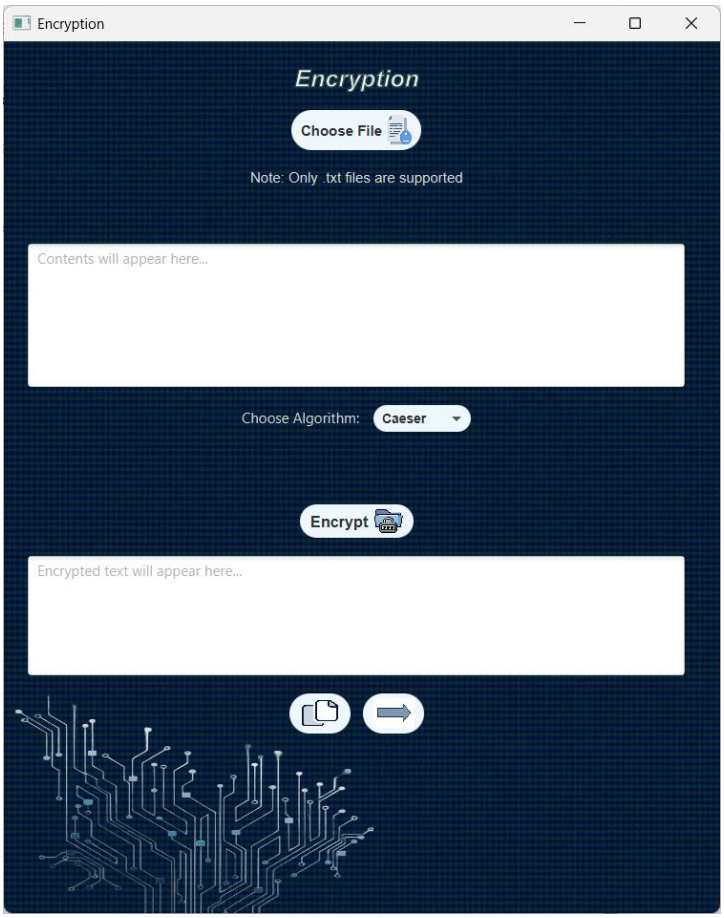
- If you want to **decrypt** a file, proceed to **Step 3B**.

3.2 Encrypting a File

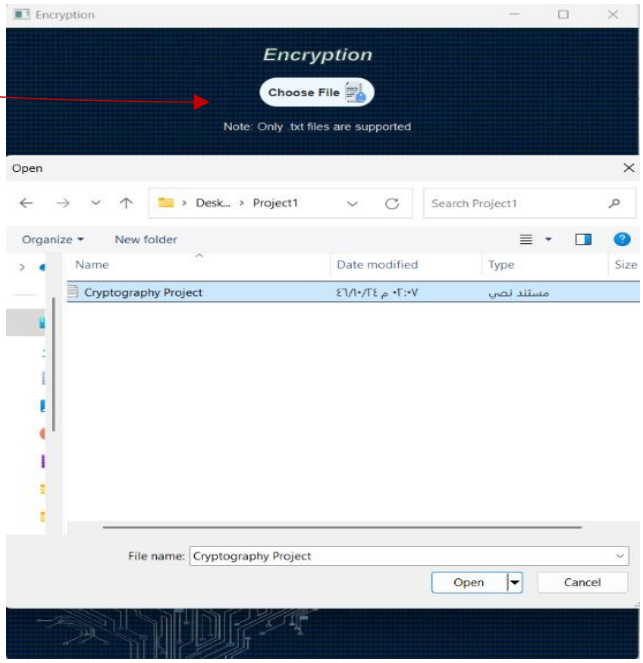
Users can encrypt a file by following the steps below:

Step 3A: Encrypting a Text File

❖ Click the **'Encryption'** button on the selection screen.
❖ The encryption interface will appear, as shown.



- Click **Choose File** and select a `.txt` file to encrypt.



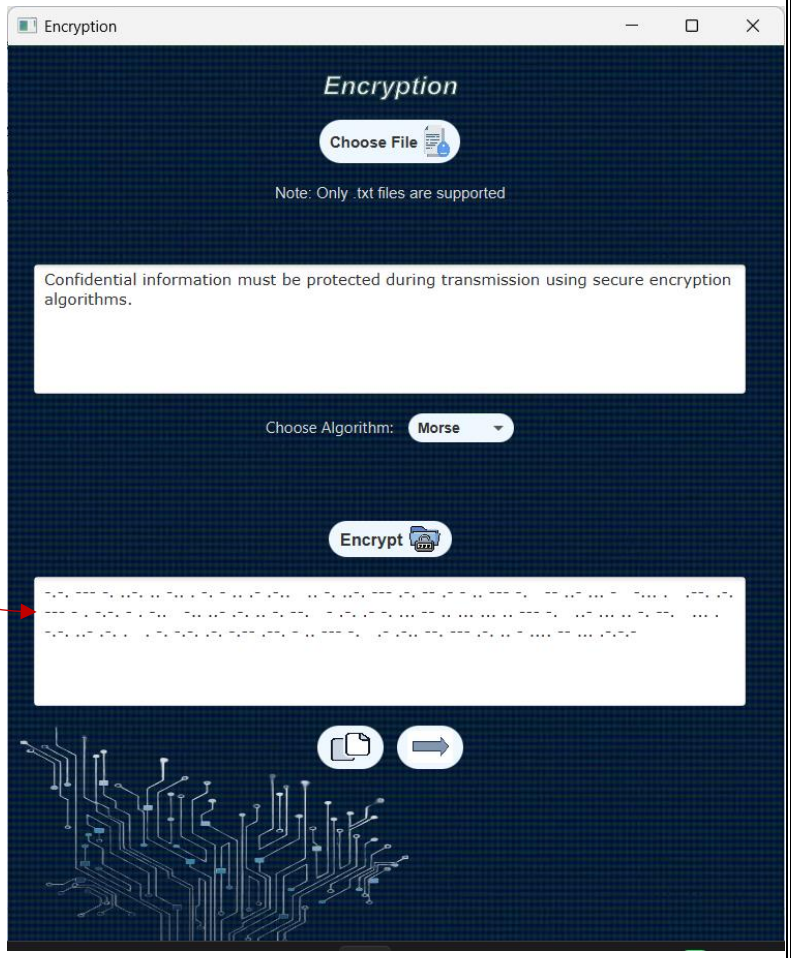
- The contents of the selected file will be displayed in the input text area.

- Select an encryption algorithm from the dropdown menu (e.g., *DES*).
- If the algorithm requires a key, enter it in the corresponding field.

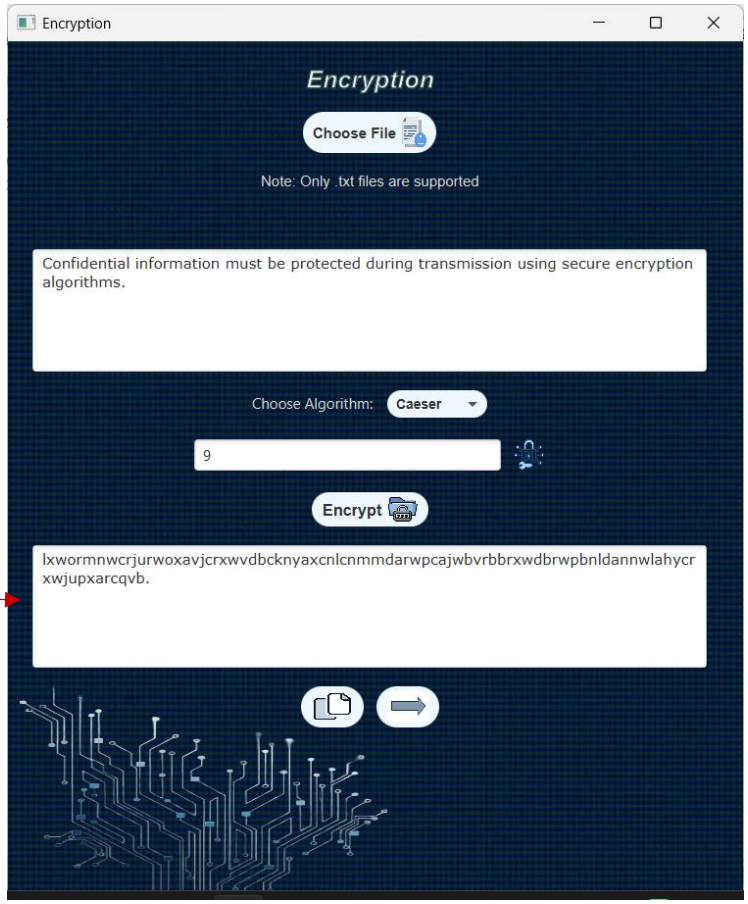
- Click the **Encrypt** button.

- The encrypted result will appear in the output area below.

- Encryption using Morse



- Encryption using Ceasar



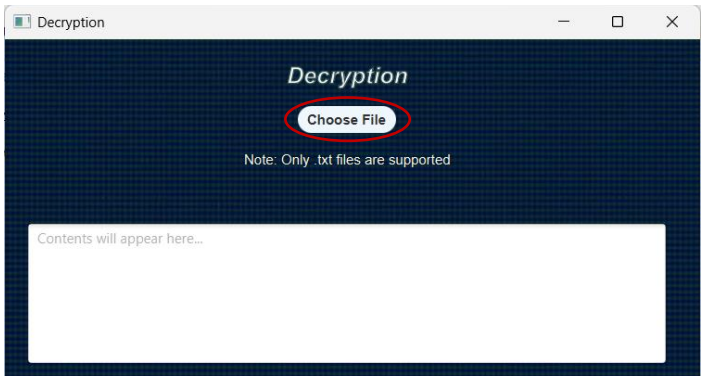
3.3 Decrypting a File

Users can decrypt a previously encrypted file by following the steps below:

Step 3B: Decrypting a Text File

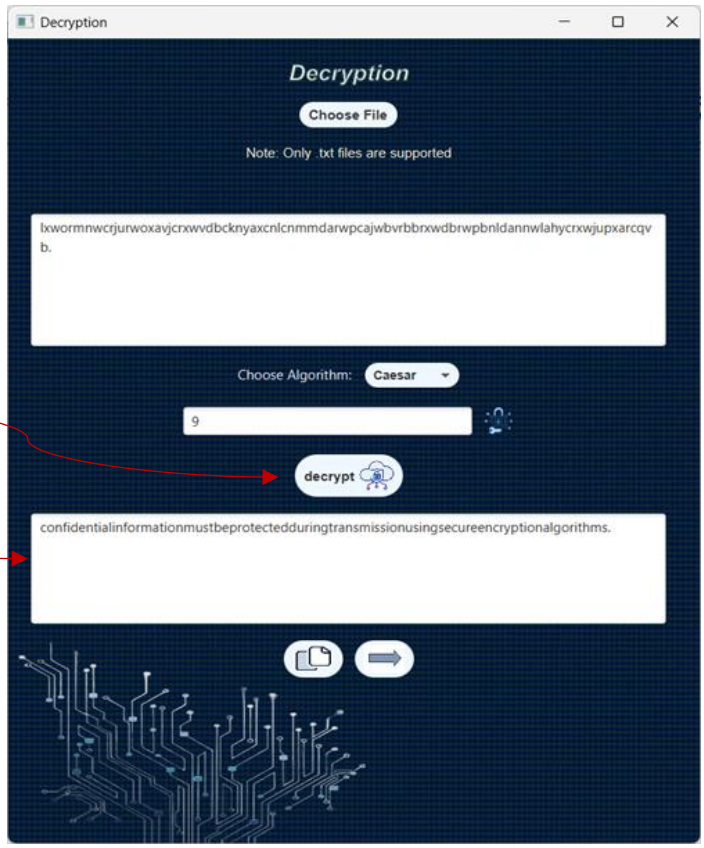
- Click the **"Decryption"** button from the selection screen to access the decryption interface.

- Click **"Choose File"** to upload a .txt file containing the encrypted text.
  Note: You can also paste or manually enter the encrypted text into the input field.

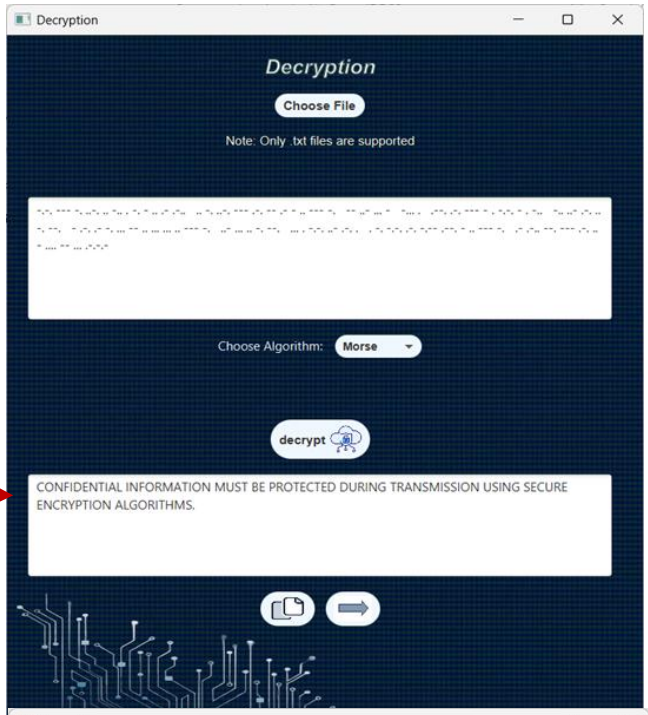


- Select an encryption algorithm from the dropdown menu (e.g., *DES*).
- If the algorithm requires a key, enter it in the corresponding field.

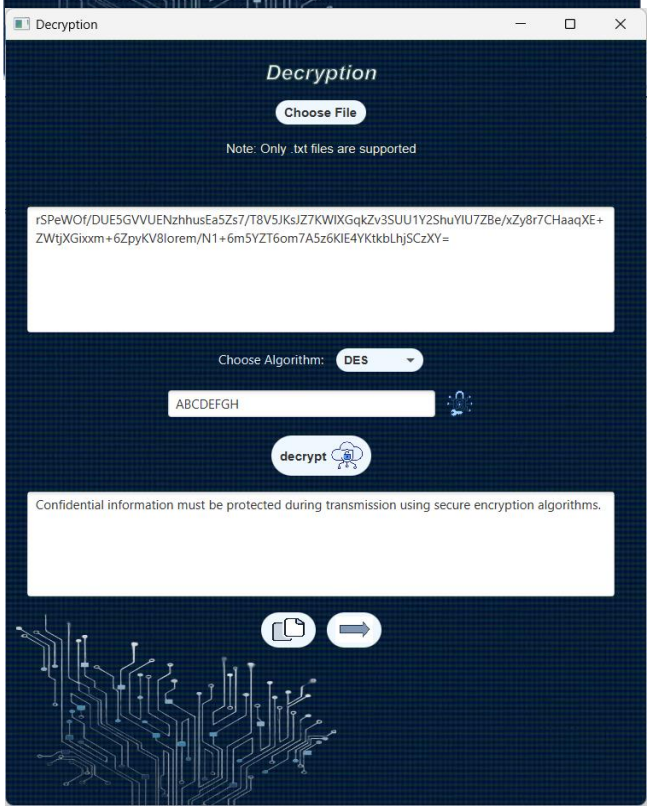
- Click the **Decrypt** button.
- The plaintext will be displayed in the output area.



❖ Decryption using Morse



❖ Decryption using DES



4. Encryption Techniques Used

4.1 Caesar Cipher

The Caesar Cipher algorithm is one of the oldest and simplest encryption methods. It is attributed to Julius Caesar, who used it to protect his military messages. The algorithm relies on moving each letter in the text a fixed number of positions in the alphabet, known as the key. [3]

Encryption/Decryption:
In the encryption process, each letter in the plaintext is replaced by another letter obtained by moving it by the key. [4]
To decrypt, the process is reversed by moving the letters by the same number of positions but in the opposite direction. [5]

The mathematical formula for encryption:
$E(x) = (x + k) \bmod 26$
Example:
C = 2, and the key k = 3
$E(2) = (2 + 3) \bmod 26 = 5 \rightarrow 5 = F$
The mathematical formula for decryption:
$D(x) = (x - k) \bmod 26$
Example:
F = 5, and the key k = 3
$D(5) = (5 - 3) \bmod 26 = 2 \rightarrow 2 = C$

4.2 Morse Code

Morse Code is a substitution encoding system that represents letters, numbers, and punctuation marks using a sequence of dots (·) and dashes (–). It was originally developed in the 1830s and 1840s for telecommunication using telegraph lines.

Morse Code represents letters and numbers using dots (·) and dashes (–). These signals can be transmitted via:

- **Sound** – such as short and long beeps or taps.
- **Light** – such as short and long flashes with a flashlight.
- **Electrical signals** – like those used in early telegraph machines

Here's a link that demonstrates how Morse Code is used through sound:
https://youtube.com/shorts/GJ6r_T6ckc4?feature=shared

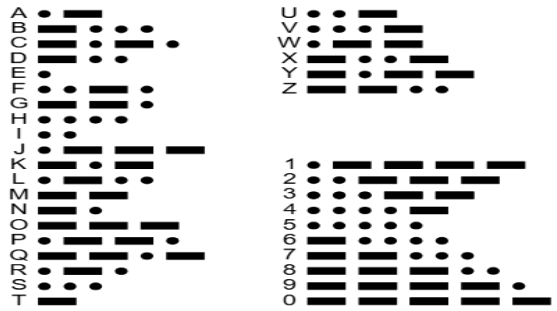
One of the most famous examples of using Morse Code:
During the Vietnam War, specifically in 1965, U.S. Navy Commander Jeremiah Denton was captured after his aircraft was shot down over North Vietnam. During his captivity, he was subjected to severe torture and harsh detention conditions that lasted for nearly eight years, four of which he spent in solitary confinement.

In 1966, Denton was forced to appear in a propaganda television interview organized by the North Vietnamese government, aiming to show that American prisoners of war were being treated well. However, during the interview, Denton cleverly seized the opportunity to send a secret message to U.S. intelligence using Morse Code. He blinked

his eyes in a specific pattern to spell out the word "TORTURE", confirming that prisoners were being mistreated.[7][8]

You can watch Jeremiah's interview through this link
https://youtube.com/shorts/8JRImSZqklc?feature=shared

example:



the word "**HELLO**" in Morse code becomes:

HELLO → • • • • • • – • • • – • • – – –

Modern Uses of Morse Code:

- Amateur Radio (Ham Radio): Used for long-distance communication in weak signal conditions, due to its simplicity and low power needs.
- Emergency Situations: Serves as a backup communication method using light or sound signals, with SOS (··· —— ···) being universally recognized.
- Aviation and Maritime Navigation: Still used for identifying stations in aviation and maritime communication.
- Training and Education: Taught in military training for emergency communication in remote or challenging conditions.

4.3 DES (Data Encryption Standard)

DES (Data Encryption Standard) is a kind of symmetric-key block cipher developed by a team at IBM in the early 1970s used for encrypting and decrypting data. It was used to protect sensitive information. Being a symmetric algorithm means it uses the same key for both encryption and decryption. The DES algorithm processes the input text in blocks of 64 bits and transforms it into encrypted text using keys of 56 bits. [8]

How does DES work?

DES uses 64-bit blocks of data encryption while the key used is set at 56 bits. Each set of DES encryption consists of an initial step of permutation followed by 16 rounds of encryption.
The rounds are performed using a Feistel network algorithmic structure and subkeys via substitution, permutation, and bitwise XOR. The subkeys are generated from the original key. After the rounds are completed, a final permutation is applied to produce the encrypted output (ciphertext). [9]

Example:

1. The sender and receiver agree on a shared key, for example: (password)
2. The sender uses this key to encrypt the message, for example: (this is DES).
3. The message is converted into binary and divided into two halves.
4. Sixteen subkeys are generated to go through 16 rounds. In each round, the bits are rearranged and combined with a subkey using specific functions.
5. A final permutation is applied, and the ciphertext appears, for example: (vlAm1Ae0QqmFO70vJZHW+g==).
6. The receiver uses the same key (password) to decrypt the ciphertext by applying the same steps in reverse to recover the original message: (this is DES). [10]

5. Troubleshooting Guide

Issues	Solutions
Invalid file format	You must ensure that the file type is a text file
JavaFX not displaying UI	The JavaFX library must be properly configured, and the path should be correct.
Invalid Caesar shift value	In Caesar cipher encryption, a shift value must be selected between 0 and 25.
Algorithm may fail if inputs don't match its requirements.	Ensure inputs align with the selected algorithm.

6. Additional Resources:

Algorithm	Resources
Ceasar	GeeksforGeeks. (n.d.). *Caesar Cipher in Cryptography*. GeeksforGeeks. https://www.geeksforgeeks.org/caesar-cipher-in-cryptography/ codedamn. (2023, March 15). *Caesar Cipher: Introduction*. Codedamn. https://codedamn.com/news/caesar-cipher-introduction 101 Computing. (n.d.). *Caesar Cipher*. 101Computing. https://www.101computing.net/caesar-cipher/
DES	GeeksforGeeks. (n.d.). *An accessible explanation of DES's working principles and its significance in cryptography*. https://www.geeksforgeeks.org/data-encryption-standard-des-set-1/ NIST. (2001). *An official overview of DES's development and its role in cryptographic standards*. https://csrc.nist.gov/pubs/book-section/2001/01/data-encryption-standard/final Simplilearn. (n.d.). *Data Encryption Standard (DES) Algorithm in Cryptography*. https://www.simplilearn.com/what-is-des-article Zenarmor. (n.d.). *What is Data Encryption Standard (DES)?* https://www.zenarmor.com/docs/network-security-tutorials/what-is-data-encryption-standard-des *(DES)*. In *Contemporary Cryptology* (pp. 55–80). Springer. https://link.springer.com/chapter/10.1007/978-1-4615-0043-8_2
Morse	Morse Code Translator. (n.d.). *Morse Code World*. Retrieved April 30, 2025, from https://morsecode.world/international/translator.html Britannica Kids. (n.d.). *Morse code*. Encyclopedia Britannica. Retrieved April 30, 2025, from https://kids.britannica.com/kids/article/Morse-Code/390814 The Tech Guy. (2021, February 18). *What is Morse Code and how to use it* [Video]. YouTube. https://www.youtube.com/watch?v=D8tPkb98Fkk

7. Conclusion

7.1 Summary

In this project, a tool was developed to encrypt and decrypt text files using the Caesar cipher, Morse code, and DES algorithms. The goal was to develop an easy-to-use tool for encrypting and decrypting text. This tool helps users understand how encryption works in practice. Through practical application, we gained a deeper understanding of how encryption techniques work in digital security and data protection using various techniques. This tool gives users the ability to encrypt their messages or text files to protect their privacy and prevent unauthorized access.

7.2 Achievements

1. Develop a practical tool for encrypting and decrypting text.
2. Integrate three different encryption algorithms into a single tool.
3. Develop an intuitive interface that allows users to easily experiment with encryption.
4. Gain practical knowledge about data security and protection methods.

8. References

Section	References
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Caesar Cipher	3. GeeksforGeeks. (n.d.). *Caesar cipher in cryptography*. Retrieved April 30, 2025, from https://www.geeksforgeeks.org/caesar-cipher-in-cryptography/ 4. codedamn. (2023, March 15). *Caesar cipher: Introduction*. Retrieved April 30, 2025, from https://codedamn.com/news/caesar-cipher-introduction 5. 101 Computing. (n.d.). *Caesar cipher*. Retrieved April 30, 2025, from https://www.101computing.net/caesar-cipher/
Morse	6. Almond Solutions. (n.d.). *What are the modern uses of Morse code?* Retrieved April 30, 2025, from https://www.almondsolutions.com/blog/what-are-the-modern-uses-of-morse-code 7. wikiHow. (n.d.). *How to learn Morse code*. Retrieved April 30, 2025, from https://www.wikihow.com/Learn-Morse-Code
DES	8. National Institute of Standards and Technology. (n.d.). *Data encryption standard overview* (NIST SP 958). Retrieved April 30, 2025, from https://nvlpubs.nist.gov/nistpubs/sp958-lide/250-253.pdf 9. Sahu, R. (n.d.). *A comprehensive guide to the data encryption standard (DES) algorithm in cryptography*. Medium. Retrieved April 30, 2025, from https://medium.com/@riteshs4hu/a-comprehensive-guide-to-the-data-encryption-standard-des-algorithm-in-cryptography-7e29a3cbfd05 10. TutorialsPoint. (n.d.). *Data encryption standard (DES)*. Retrieved April 30, 2025, from https://www.tutorialspoint.com/cryptography/data_encryption_standard.htm